Institute/ School Name	School of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Program Name	Bachelor of Engineering (Computer Science & Engineering): B.E (CSE)		
Course Code	25CS031	Course Name	Cloud Computing Essentials
L-T-P (Per Week)	3-0-4	Course Credits	05
Academic Year	2025-26	Semester/Batch	6th /2023-27
Course Coordinator	Dr. Ravita Chahar		

1. Course Outline:

Fundamentals of Cloud Computing, AWS Compute Services, AWS Storage Services, AWS Database Services, Network and Content Distribution, Identity Management, AWS Containers Service, Introduction to Dockers and Kubernetes, Security, Identity, and Compliance, AWS Management and Governance, AWS Cost Management, Developer Tools.

2. Programme Outcomes (POs):

At the end of the programme, students will be able to:	
PO 1	**Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	**Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	**Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

3. Course Learning Outcomes (CLO):

After completing the course, the students will be able to:

CLO1: Explain fundamental cloud computing concepts, deployment and service models, cost, compliance, and legal aspects to build strong conceptual understanding and foundational cloud skills.
CLO2: Apply virtualization, compute, and storage skills by creating and managing AWS resources such as EC2, AMIs, auto-scaling groups, load balancers, and storage services (S3, EBS, Glacier) following best practices.
CLO3: Develop cloud networking and security skills by designing VPCs, configuring IAM, security groups, encryption, and ensuring compliance and business continuity requirements.
CLO4: Evaluate cloud tools and deployment models (Docker, Kubernetes, CloudFormation, CloudWatch) to design scalable and cost-effective solutions that enhance professional competence and employability.
CLO5: Design cloud-native applications using AWS databases, analytics, and Generative AI services to build intelligent solutions that strengthen innovation capability and industry employability.

4. CLO-PO Mapping Matrix:

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO1	H	M	-	-	L	-	-	-	-	-	L	L
CLO2	M	M	H	M	H	-	-	-	L	L	M	L
CLO3	H	H	H	M	H	L	-	-	L	L	M	M
CLO4	M	M	H	M	H	-	-	-	L	L	H	M
CLO5	H	H	H	M	H	-	-	-	M	M	M	H

5. ERISE Grid Mapping:

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	3
Research/Innovation	1
Skills	5
Employability	5

6. Recommended Books (Reference Books/Textbooks):

B01: Cloud Computing: Concepts, Technology & Architecture by Erl, (1st Edition), Pearson Education India
B02: AWS Certified SysOps Administrator Official Study Guide: Associate Exam by Stephen Cole, ISBN No – 978- 1- 119-37742-9.
B03: AWS Certified Solutions Architect Official Study Guide (1st Edition), Joe Baron.
B04: AWS Security: Best Practices on AWS by Albert Anthony (1st Edition), Ingram short title.
B05: AWS Certified Security – Specialty Study Guide by Ben Piper & David Clinton (1st Edition), Sybex.

7. Other readings and relevant websites:

Resources	Link of Journals, Magazines, Websites and Research Papers
R1	https://www.amazon.com/Certified-SysOps-Administrator-Official-Study/dp/1119377420
R2	https://www.amazon.in/Certified-Solutions-Architect-Official-reprint
R3	https://www.qwiklabs.com/home?locale=en
R4	https://docs.aws.amazon.com/index.html#lang/en_us
Resources	**Link of Audio-Video resources**
V1	https://www.youtube.com/watch?v=HbIMwaWvT6g
V2	https://www.youtube.com/watch?v=r6Q_QfmYpLQ

* Resources uploaded on ERP system is accessible to all the students registered for the course.

8. Recommended Tools and Platforms:

- AWS Free tier Account
- VMware Workstation Player 16

9. Course Plan:

Lecture Number	Topics	Weightage in ETE (%)	Instructional Resources
1-4	Cloud Computing Basics – History of Cloud Computing, Characteristics of Cloud Computing, Need for Cloud computing, Advantages and Possible Disadvantages of cloud computing	30	B01, R4
5-6	Cloud Deployment Models – Public, Private, Hybrid, Community, Cloud Service Models: IaaS, PaaS, SaaS		
7-8	Hypervisors, Type I and II, Creating Virtual Machine using VMware Player or Hyper V, Introduction to AWS portal		
9-12	Introduction to EC2, Instance Types and Uses, Amazon Machine Images (AMI's),		B03, R3, R1, R2
13-15	Auto scaling Instances, Modifying Existing Images, Creating New Images of Running Instances, Instances Backed by Storage Types		
16-21	Elastic IPs, Elastic Load Balancing		
22-24	Cloud data lifecycle phases, cloud data storage types, Amazon Storage, S3 Storage Basics		
25-28	Buckets and Objects, Managing Voluminous Information with EBS, Glacier Storage Service		
29-32	Describe Amazon Dynamo, understand key aspects of Amazon RDS, Launch an Amazon RDS instance		
33-36	Introduction to AWS Networking, Setting Up a Security Group, Access Control Lists (ACLs)		
37-39	Setting Up VPC And Internet Gateway, Cloud front, For VPC Scenario 1: VPC With a Public Subnet Only (Standalone Web),	30	B03, R3, V1
40-42	Scenario 2: VPC with Public and Private Subnets (3 Tier App)		
43- 45	Introduction to AWS Security, Amazon Identity and Access Management (IAM)		
46-47	Introduction to containers, Dockers and Kubernetes,		
48-49	how to create, execute, change, and deploy containers. container management		
50-51	Introduction to Elastic Beanstalk	20	B03, R3, V2
52-54	Infrastructure as Code: Learn to manage and provision infrastructure as code using AWS CloudFormation		
55-59	Introduction to Cloud Watch, Describe Amazon Cloud Watch metrics and alarms, AWS Messaging Services (SNS).		
60-62	AWS Directory Service, AWS Key Management Service		
63-65	Use AWS encryption solutions to protect data at rest, Compliance Issues: Data privacy, Data security, Contracts, Regulations	10	B04, R2
66-71	Cloud Cost Models: Pay-As-You-Go, Prepaid/Fixed Subscriptions, Reserved Instances, AWS Savings Plan, Spot Instances, Cloud computing resources, Cloud Dos and Don'ts. Business continuity technology solutions - Data replication, Automated failover systems, Scalable recovery solutions		B04, B05
72-75	Legal challenges involved in Cloud- Data breaches, Privacy violations, Compliance, Copyright issues, Limited liability, Data confidentiality, Data governance.		B04, B05

76-79	Budgeting, Cost management, resource planning.		B04, B05, R3
80-84	Introduction to ML ecosystem of AWS & tools - Amazon Lex	10	B03, V1, R4
85-90	Introduction to AWS Analytics tools - Amazon Kinesis		
90-94	Generative AI and Agentic		
95-100	AI Services: Amazon Bedrock AgentCore, Amazon S3 Vectors		
101-105	Kiro IDE and AI-Driven Development Lifecycle (AI-DLC)		

10. Industry Interventions:

- The students will receive resources on the LMS system, and their progress will be tracked on the platform based on their submissions (number of submissions, errors encountered, and optimized solutions).

11. Action plan for different types of learners:

Slow Learners	Average Learners	Advanced Learners
Remedial Classes	Workshops/Practice Assignment	Projects assignment

12. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
Internal Component	Sessional Tests (STs)	02*	40%	Online**
External Component	End Term Examination (ETE)	01	60%	Online**
Total		**100%**		

* Out of the two, the best 01 will be considered.
** Proctored examination will be conducted on Testpad platform.

13. Details of Evaluation Components:

Evaluation Component	Description	Syllabus Covered (%)	Timeline of Examination	Weightage (%)
Internal Component	ST 01	Up to 40% (Lectures 1-42)	Week 7	40
	ST 02	41% - 80% (Lectures 43-84)	Week 12	
External Component	End Term Examination*	100	As Notified by the Exam Cell	60
Total				100%

* Minimum 75% attendance is required to become eligible for appearing in the End Term Examination

14. Format of Evaluation Components:

Type of Assessment	Total Marks	1 Mark MCQ	2 Marks MCQ	5 Marks Coding	10 Marks Coding
Sessional Tests	40	10	5	2	1
End Term Examination	60	10	10	2	2

15. Revision (if any):

Academic Yearof Previous Version	2024-2025	**Percentage of Revision**	11%
Topics:			
Added: • Generative AI and Agentic AI Services: Amazon Bedrock AgentCore, Amazon S3 Vectors • Kiro IDE and AI-Driven Development Lifecycle (AI-DLC)			

16. This Document is:

Designation	**Name**	**Signature**
Prepared by Course Coordinator	Dr. Ravita Chahar	
Verified by Assistant Dean	Dr. Ashutosh Kumar Dubey | Dr. Ravita Chahar	
Date	January 10, 2026	